The mathematics first

Exactly four point shields are necessary and sufficient. In the side-4 portal square, $S = (0, 0)$ and $T = (2, 2)$. Opposite edges are joined by translation: a shot keeps its direction. Every shot that reaches T lifts to a straight segment from the origin to a target copy $(2 + 4m, 2 + 4n)$, for integers m, n.

The midpoint of the segment ending at its *first* target hit folds to one of

$$(1, 1), \quad (1, 3), \quad (3, 1), \quad (3, 3).$$

These four shields therefore intercept every relevant shot before T. Conversely, the four short diagonal paths have disjoint interiors, so no single point can block two of them. Three shields cannot suffice.

Assumptions that matter. A shield is an exact point, never S or T; a pencil cross marks it, and it has no blocking radius. Repeated copies represent the same shield. Travel is continuous along straight lines, not restricted to grid roads. Shots stop at their first target hit. A ray that never reaches T is outside the blocking question.

What children discover. Four diagonals give a lower bound, and two drawn portal shots force two of the shields; unsuccessful shield arrangements lead to a better arrangement; repeated-room pictures turn wraparound into straight lines; finally, a midpoint argument upgrades tests to an all-shot explanation. Testing the finite map is evidence for a conjecture, not its proof.

Entry and depth. This shared Grades 4–5 prototype is for children who can locate grid points, match a point across room copies, and trace a straight line. The final explanation adds halves, even/odd patterns and reasoning about all possibilities. Formal coordinates are optional in a child's explanation. It is not a tested age-placement claim.

Set up, launch, and a first route

Per pair: sharp pencil, eraser, straightedge, the four student pages, spare page 2–3 maps or erasable sleeves, and a piece of tracing paper about 6 cm square. Print at actual size. Children mark shields with small pencil crosses; if you want objects, use clear counters with a centre dot, never opaque ones, which hide the line exactly where it must be checked. On page 3 a child traces the bold room's outline and the shield crosses onto the tracing paper, then lines the outline up with any room copy a test line crosses: the crosses land on that copy's shields, so the paper keeps the repeat rule. Keep room edges and grid lines visible. At the older table, two children investigate and the third acts as rotating shot-checker; an adult helps with accurate transfers.

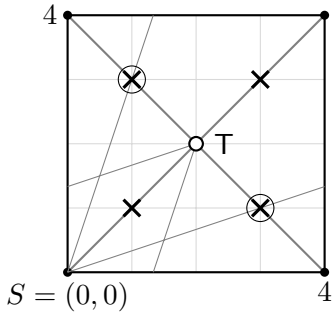
Common launch, about 5 minutes. Let everyone trace an edge crossing with a finger and mark a shield with a small cross. Use the printed non-task example: from $(3, 1)$, the shot crosses the right seam at $(4, 1\frac{1}{2})$ and meets the copy $(5, 2)$ of shield $P = (1, 2)$. In one room it reappears at $(0, 1\frac{1}{2})$ and reaches P. Have children identify the two drawings of one shield, then swap tracer and checker. Do not reveal the four universal positions. Show the tracing square when a pair reaches Problem 3.

Manageable first meeting: spend about 5 minutes on Problem 1, then 10–15 on the four contrasting shots in Problem 2. Use the remaining exploration time on Problem 3 and share one improved arrangement. Stop with a checked four-shield conjecture. Return to Problem 4 when the group wants a guarantee; finishing every page is unnecessary. A possible Grades 2–3 route, untested, about 15–20 minutes: if the whole circle uses this week, do Problem 1 (finish four lines, then block all six with the fewest crosses), which needs no coordinates once an adult has traced the two drawn shots through their edges, then Problem 2's shots A–C, laying a ruler from S to each letter and marking the first T reached: A and B are Problem 1's drawn shots unfolded; C stops at the room's own T.

Rehearse before use. Check that the traced outline lines up with the page-3 room copies when printed at actual size, a ruler can reach the target copies, and children can preserve height at a seam. No physical setup rehearsal or classroom pilot has occurred.

Answers and optional hints

Problem 1. The minimum is **4**. The four corner-to-centre diagonals share no point except S and T, so each needs its own shield. The steep drawn shot (toward the copy $(2,6)$: $(0,0)$ to $(\frac{4}{3},4)$, then $(\frac{4}{3},0)$ to T) meets the diagonals only at $(1,3)$, and the shallow one (toward $(6,2)$) only at $(3,1)$. So a four-shield answer must use $(1,3)$ and $(3,1)$; the other two shields go anywhere strictly inside the diagonals from the lower-left and upper-right corners. A child who puts one shield where the two drawn shots cross, at $(\frac{1}{2},1\frac{1}{2})$ or $(1\frac{1}{2},\frac{1}{2})$, needs five; that is a good arrangement to improve.



The circled shields are forced; the other two shown are one choice, and they also solve the later problems.

Hint if needed: "Where does a drawn shot cross a diagonal?" The diagonals' only shared endpoints are forbidden shield locations. A disk that appears to cover two lines is not a legal point-shield explanation.

Problem 2. The first target depends on the whole segment, not the target name chosen by the shooter.

Aimed-at copy	First target reached	Halfway shield, folded into room
A $(2,6)$	$(2,6)$	$(1,3)$
B $(6,2)$	$(6,2)$	$(3,1)$
C $(6,6)$	$(2,2)$	$(1,1)$
D $(10,6)$	$(10,6)$	$(1,3)$

A simple legal transfer uses $(2,6)$, the steep shot of Problem 1: draw from $(0,0)$ to $(\frac{4}{3},4)$, reappear at $(\frac{4}{3},0)$, then continue in the same direction to $(2,2)$. The shot to $(6,2)$ is the sideways version. The shot toward $(6,6)$ stops before any portal crossing. The shot to $(10,6)$ crosses right, top, then right before its first T.

Hint if needed: cover the part of the ruler line beyond the first target. For $(6,6)$, finding a shield halfway to the *chosen* distant copy would be too late. In the table, every midpoint is halfway to the *first* hit.

Problem 3. The four positions $(1,1),(1,3),(3,1),(3,3)$ work, and they are the only four-shield answer on the map: Problem 1's shots force $(1,3)$ and $(3,1)$, the shot toward $(6,-2)$ meets the diagonals only at $(3,3)$, and the shot toward $(10,2)$ then forces $(1,1)$. Shields go in the bold room; the tracing square carries them to every room copy a tested line crosses. For example the halfway point toward $(10,6)$ is $(5,3)$, which is the copy of shield $(1,3)$ one room to the right.

The map contains 16 target copies, with each coordinate in $\{-2,2,6,10\}$. Every first-target segment to these copies is intercepted by the arrangement. Let partners pool tests rather than require every child to duplicate every line. A successful finite test does not yet explain shots outside the map.

Hint sequence: "Can your partner find an escaping line?" Then, if needed, "Try points at equal fractions along the four short diagonals." Reserve "halfway" for a pair that wants a stronger hint.

Why four work for every shot

Problem 4. The same four shields suffice for arbitrarily long shots. Three can never suffice.

Sufficiency. Unfold the journey only as far as the first target hit. That endpoint is some $(2+4m, 2+4n)$, even if it is far outside the printed window. The point halfway there is

$$\left(\frac{2+4m}{2}, \frac{2+4n}{2}\right) = (1+2m, 1+2n).$$

In each coordinate, reducing modulo 4 leaves either 1 or 3. The midpoint is therefore a copy of one of the four shields. It occurs strictly before the target, and cannot be either forbidden endpoint. This proves interception for *every* target-reaching direction, without listing them.

A drawing-level explanation. Each target copy has coordinates 2 more than a multiple of 4. Halving either coordinate puts it at an odd grid coordinate. Odd grid points fold to the two positions 1 and 3 in each direction, making only four possible halfway locations. Children may show this with doubled strips or several copies before using the formula.

Necessity. Look only at the four shortest shots, with displacements $(2,2)$, $(2,-2)$, $(-2,2)$ and $(-2,-2)$. Their open interiors occupy four different quadrants of the room cut at $x=2$ and $y=2$. In the printed room they are the four corner-to-center segments. These interiors never meet, even after identifying opposite edges. A shield away from S and T can lie on at most one. Every one of the four shots needs an interception, so at least four shields are necessary. This lower bound and the midpoint construction agree.

If the proof stalls: return to the four short shots for necessity before addressing sufficiency. Ask "What would one point have to do if there were only three?" For sufficiency, use a large off-map target chosen by the children and fold its halfway point; then ask what stays unchanged as the copy changes.

Extension: shots back to S. Take T away and ask how few shields stop every shot that returns to S. Unfolded, a returning shot first reaches an S copy $(4a, 4b)$ with a and b sharing no factor, so they are not both even. Its halfway point $(2a, 2b)$ folds to $(2,0)$, $(0,2)$ or $(2,2)$, so these three shields stop every returning shot; $(2,0)$ is the same point as $(2,4)$. The shots to $(4,0)$, $(0,4)$ and $(4,4)$ run along the bottom edge, the left edge and the diagonal and share no point except S, so three are needed.

Use the conclusion carefully

What is established: the exact minimum for this point-source, point-target portal model. The proof does not concern positive-size obstacles or mirror bounces. Different target/source positions need their own argument; do not carry over the four-diagonal lower bound without checking it.

What to notice in a first use: can children distinguish one shield from its copies, stop at an earlier target, and check that a line passes exactly through a cross? Record which representation needed adult help and whether the finite testing game held their interest. These are observations to gather, not results already known.

Source and verification

Samuel Lelièvre, Thierry Monteil and Barak Weiss, *Everything is illuminated*, arXiv:1407.2975, opening blocking-set definition (p. 1) and Lemma 12(a)–(c) (pp. 12–13). The source gives torus midpoint blocking and blocking cardinality four. The side-4 boards, chosen shots and elementary diagonal lower bound here are authored specializations.

The accompanying independent standard-library checker verifies the four task rays, the six Problem 1 shots, all 16 printed target copies, wider samples, the returning-shot extension and the launch coordinates. Finite computation supports the printed answers; the arguments above establish the universal claims. This guide matches the four-page GGT70-S-v2 student packet. Digitally checked, unpiloted; physical rehearsal remains outstanding.